CHAPTER 1: INTRODUCTION AND PROBLEM IDENTIFICATION

1.1 Introduction

Cybersecurity has become an essential part of modern information technology because organizations depend on connected systems, applications, networks and digital services. As threats become more complex, organizations need practical ways to test defensive controls, understand attack behaviour and improve incident response. Cyber ranges provide controlled environments in which security scenarios can be exercised without directly exposing production infrastructure.

Artificial intelligence and multi-agent systems can support security analysis, orchestration and decision support. HavenSec Arena applies these ideas to a controlled cyber-resilience simulation and proposes an AI-assisted multi-agent Red Team versus Blue Team platform.

1.2 Background of the Project

Cyber ranges are interactive simulated environments that replicate networks, systems, tools and applications for cybersecurity education, training and security testing. NIST's *The Cyber Range: A Guide* describes cyber ranges as safe and legal environments for hands-on cyber skills and explains simulation and emulation approaches.

Existing technologies already provide important parts of this area. MITRE CALDERA automates adversary emulation using ATT&CK; Atomic Red Team provides focused ATT&CK-mapped; tests; Cyberbit provides cyber-range exercises and readiness measurement; and Immersive Labs provides collaborative cyber-range exercises.

1.3 Problem Identification

The identified problem is the lack of a single controlled workflow that combines simulated offensive activity, defensive monitoring and response, objective scoring and post-exercise improvement feedback while keeping all activity inside an isolated environment.

- **What is the problem?** Cyber-resilience exercises can be difficult to run repeatedly and objectively measure.
- **Who faces this problem?** Organizations and security teams that need repeatable readiness testing.
- **Where does it occur?** During security exercises, incident-response drills and defensive readiness assessments.
- **Why does it occur?** Production systems cannot safely be used as unrestricted attack-training environments, while manual exercises require significant coordination.
- **What are the consequences?** Limited practice, inconsistent telemetry, difficult comparison of results and weak visibility into improvement areas.

1.4 Problem Statement

“To design and develop a safe, isolated and AI-assisted cyber-resilience simulation platform for controlled Red Team and Blue Team exercises in order to overcome the safety, repeatability and objective-scoring limitations of practicing on real systems.”

1.5 Need for the Proposed Solution

HavenSec Arena is required to provide a controlled environment in which security scenarios can be repeated, observed and measured without depending on live production targets. The NIST cyber-range guidance highlights realism, legality, flexibility, accessibility and scalability as important considerations.

- **Existing limitations:** Real production infrastructure is unsuitable for unrestricted attack simulation.
- **User difficulties:** Teams need repeatable scenarios, clear event timelines and measurable outcomes.
- **Cost:** Dedicated physical training infrastructure can be expensive; software isolation can reduce infrastructure requirements for a prototype.
- **Efficiency:** Automated orchestration can coordinate simulation, telemetry, detection, response and scoring.
- **Accuracy:** Defined metrics can make results more consistent across repeated exercises.
- **Accessibility:** A software prototype can make controlled exercises easier to reproduce.
- **Industrial requirement:** Organizations need to validate readiness without risking production systems.

1.6 Objectives

- 1. To study existing cyber-range, adversary-emulation and defensive-training systems.
- 2. To identify limitations of existing solutions for combined simulation, defense and scoring.
- 3. To design an isolated cyber-range concept using dummy data and simulated events.
- 4. To design a multi-agent workflow containing Red Team, Blue Team and Referee roles.
- 5. To generate an incident timeline and a composite cyber-resilience score.
- 6. To evaluate the proposed workflow and identify future improvements.

1.7 Scope of the Project

Included in the project

- A controlled sandbox/cyber-range concept using dummy or synthetic data.
- Red Team simulation, Blue Team monitoring/response and Referee scoring.
- AI-assisted agent coordination and an incident timeline/reporting workflow.

Not included in the project

- Attacking or testing real production systems, public targets or unauthorized infrastructure.
- Use of real credentials, confidential organizational data or uncontrolled external network access.
- Unrestricted autonomous offensive activity outside the sandbox.

1.8 Limitations

- The prototype is limited to controlled simulated scenarios and does not represent the full complexity of a production Security Operations Center.
- The sandbox uses synthetic/dummy data rather than live organizational telemetry.
- The quality of the resilience score depends on the event data and metrics available to the Referee component.
- The number and realism of scenarios are limited by project time, hardware and development resources.
- Integration with enterprise SIEM, EDR or other operational tools is outside the initial prototype scope.
- AI-generated decisions require validation and should not be treated as autonomous authorization for real-world offensive activity.

Important technology clarification

The project presentation identifies IBM Bob as the proposed reasoning core. Current IBM documentation describes IBM Bob as an AI-powered software-development/SDLC partner with agent and subagent capabilities, not as a dedicated cyber-range reasoning engine. Therefore, this report treats IBM Bob as a proposed AI component or integration option rather than as an already-existing cybersecurity engine.

This distinction keeps the project technically defensible: HavenSec Arena is the proposed system, while IBM Bob is an external AI technology that may be evaluated or integrated for selected orchestration or reasoning tasks.

CHAPTER 2: LITERATURE SURVEY AND EXISTING SYSTEMS

2.1 Introduction

The study covers cyber-range architecture, adversary emulation, ATT&CK-based; testing, team-based cyber exercises and automated security testing. Official documentation and industry sources were reviewed to identify existing systems, their capabilities and the implementation gap addressed by HavenSec Arena.

2.2 Literature Survey

Sr. No.	Author / Source	Year	Title / Source	Technology / Method	Major Finding	Limitation
1	NIST	2023	The Cyber Range: A Guide	Cyber-range simulation, emulation and virtual environments	Cyber ranges provide safe, legal and interactive environments for training and security testing.	Realism, scalability and infrastructure choices vary by implementation.
2	MITRE	2025	CALDERA	ATT&CK-based; automated adversary emulation	Automates adversary behaviour and security assessment using ATT&CK; models and decision logic.	Primarily adversary emulation rather than a complete Red/Blue/Referee learning loop.
3	Atomic Red Team	n.d.	Atomic Red Team Documentation	ATT&CK-mapped; atomic tests and execution frameworks	Provides focused tests that teams can execute to evaluate detection coverage.	Testing library/framework rather than a complete cyber-range with integrated team scoring.
4	Cyberbit	n.d.	Cyberbit Cyber Range	Virtual enterprise networks, SOC tools, attack scenarios and team exercises	Provides realistic cyber exercises, Red/Blue team training and performance measurement.	Commercial platform with vendor-controlled architecture.
5	Immersive Labs	2025	Cyber Range Exercises	Multiplayer simulated incidents and team exercises	Supports realistic team simulations, performance measurement and skill-gap identification.	Training platform rather than the proposed AI-agent architecture.

2.3 Existing System 1 – MITRE CALDERA

Working: CALDERA is an automated adversary-emulation system that uses MITRE ATT&CK; to model and reproduce adversary behaviour. MITRE describes an architecture involving adversary representation, ATT&CK; profiles, a decision engine and an agent.

Technology: MITRE ATT&CK;, adversary profiles, decision logic and agents. **Advantages:** automation, repeatability and ATT&CK-aligned; security assessment. **Limitations:** it is primarily an adversary-emulation platform and does not provide the exact HavenSec combination of sandboxed Red Team, Blue Team, Referee scoring and improvement reporting.

2.4 Existing System 2 – Cyberbit Cyber Range

Working: Cyberbit provides cloud-based and virtual cyber-range exercises with realistic enterprise networks, attack scenarios and defensive tools. Its platform supports team exercises, Red Team/Blue Team activities and performance measurement.

Technology: virtualized enterprise networks, SOC tools, attack scenarios, telemetry and performance metrics. **Advantages:** realistic environments, scenario-based exercises, team readiness measurement and scalable range provisioning. **Limitations:** it is a commercial platform with vendor-defined architecture and is not specifically the proposed multi-agent AI sandbox.

2.5 Existing System 3 – Immersive Cyber Range Exercises

Working: Immersive Cyber Range Exercises are multiplayer exercises in which teams collaborate in realistic simulations of cybersecurity incidents and are measured on performance.

Technology: simulated incident environments, team exercises, scenario catalogues and performance reporting. **Advantages:** collaborative practice, incident investigation, response skills and identification of knowledge/skill gaps. **Limitations:** it is a training/exercise platform rather than the specific AI-coordinated Red Team–Blue Team–Referee architecture proposed by HavenSec Arena.

2.6 Limitations of Existing Systems

Existing System	Major Advantage	Major Limitation
MITRE CALDERA	Automated ATT&CK-based; adversary emulation.	Does not itself provide the full proposed defensive-response and Referee scoring loop.
Cyberbit Cyber Range	Realistic team exercises, Red/Blue training and performance measurement.	Commercial/vendor-controlled platform; not the proposed custom multi-agent architecture.
Immersive Cyber Range Exercises	Collaborative simulated incidents with performance measurement.	Training platform rather than the proposed AI-agent orchestration and scoring architecture.

2.7 Research/Implementation Gap

The reviewed systems show that important parts of HavenSec Arena already exist separately: CALDERA provides automated adversary emulation; Atomic Red Team provides focused ATT&CK-mapped; tests; Cyberbit and Immersive provide cyber-range/team exercises and measurement.

The implementation gap is therefore not the claim that no similar technology exists. Instead, HavenSec Arena aims to combine selected ideas into one academic prototype workflow: an isolated sandbox, controlled Red Team simulation, Blue Team detection/response, an AI-assisted coordination layer, Referee scoring and an improvement-oriented report.

2.8 Proposed Solution – Overview

HavenSec Arena is proposed as a Docker-isolated cyber-resilience simulation environment using dummy data and controlled events. Red Team components simulate reconnaissance, path planning and objectives; Blue Team components monitor, detect, respond and recover; and a Referee component calculates metrics such as detection time, containment, damage and recovery. The AI component is intended to assist orchestration and reasoning. This is a proposed architecture, not a claim that an identical commercial system already exists.

CHAPTER 3: ALTERNATIVE SOLUTIONS AND SOLUTION SELECTION

3.1 Introduction

Three practical approaches were considered: adopting an existing cyber-range platform, building a deterministic rule-based sandbox, and developing the proposed AI-assisted multi-agent HavenSec Arena.

3.2 Requirements of the Proposed System

❖ **Functional Requirements**

- Scenario selection and controlled scenario execution.
- Red Team simulation of predefined attack behaviours inside the sandbox.
- Generation and collection of synthetic telemetry/events.
- Blue Team monitoring, detection, response and recovery workflow.
- Referee-based measurement of detection time, containment, damage and recovery.
- Incident timeline, resilience score and improvement report generation.

❖ **Non-functional Requirements**

- Performance
- Security and isolation
- Reliability
- Usability
- Scalability
- Maintainability

3.3 Alternative Solution 1 – Use an Existing Cyber-Range Platform

Use a mature platform such as Cyberbit Cyber Range or Immersive Cyber Range Exercises. This provides established scenarios, realistic environments and exercise measurement. The major disadvantage is reduced control over the internal architecture and dependency on a third-party platform.

3.4 Alternative Solution 2 – Custom Rule-Based Cyber Sandbox

Develop a local sandbox in which predefined rules control scenario execution, event generation, detection and scoring. This approach offers strong control and is straightforward for a prototype, but it provides less adaptive reasoning and requires more custom logic for every scenario.

3.5 Alternative Solution 3 – AI-Assisted Multi-Agent HavenSec Arena

Develop a controlled sandbox with specialized Red Team, Blue Team and Referee components coordinated by an AI reasoning layer. The system is designed to combine simulation, defense, scoring and reporting while keeping all actions inside an isolated environment.

3.6 Comparison of Alternative Solutions

Criteria	Solution 1	Solution 2	Solution 3	Proposed Solution
Cost	Medium / High	Low / Medium	Medium	Medium
Accuracy	High for supported scenarios	Depends on rules	High potential; needs validation	High potential; controlled scoring
Ease of implementation	Medium	High	Medium	Medium
Hardware requirement	Medium / High	Low / Medium	Medium	Medium
Scalability	High	Medium	High potential	High potential
Security	High if properly isolated	High if properly isolated	High with sandbox controls	High with strict isolation
Maintenance	Platform dependent	High custom maintenance	Medium / High	Medium / High

3.7 Selected Solution

The selected solution is **Solution 3 – AI-Assisted Multi-Agent HavenSec Arena**, implemented as a safe, isolated cyber-resilience simulation environment.

3.8 Justification for Selection

- **Technical feasibility:** The system can be divided into sandbox, Red Team, Blue Team, Referee and dashboard/orchestration components.
- **Cost:** A prototype can use software isolation and synthetic data rather than production infrastructure.
- **Resources:** The project can use commonly available development technologies and containerization.
- **Time:** The first prototype can focus on a limited number of controlled scenarios.
- **Accuracy:** Defined metrics provide a consistent basis for comparing scenario results.
- **Availability:** Simulated scenarios can be reproduced without waiting for real incidents.
- **Safety:** The architecture explicitly confines activity to a sandbox with dummy data and no live targets.
- **Scalability:** Future versions can add scenarios, adaptive difficulty, more agents, SIEM/EDR integrations and richer reporting.

3.9 Alternative Approach

If the AI-assisted multi-agent approach does not perform as expected, the fallback will be a deterministic rule-based orchestration layer for scenario execution, event generation and scoring. AI can then be introduced incrementally for selected reasoning tasks after validation.

Conclusion of selection: HavenSec Arena is not claimed to be the first cyber-range or multi-agent cybersecurity system. Its proposed contribution is the integration of controlled simulation, defensive response, objective scoring and AI-assisted orchestration into a single academic prototype.